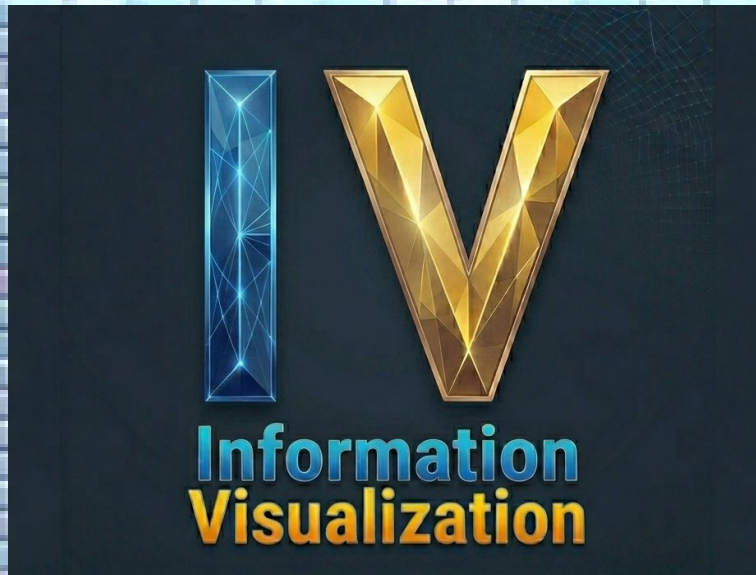


Vorlesung
Informationsvisualisierung



Kapitel 7 INTERAKTION



Lecture
Information Visualization

Chapter 7 INTERACTION

Visualization is not enough



“It is through the interactive manipulation of a visual interface – **the analytic discourse** – that knowledge is constructed, tested, refined and shared.”

Pike et al., 2009

Source: Pike, W. A., Stasko, J., Chang, R., & O’Connell, T. A. (2009). The Science of Interaction. Information Visualization, 8(4), 263–274. <https://doi.org/10.1057/ivs.2009.22>

Examples of Interactions



Chapter
7.1

Interaction Techniques

Toward a Deeper Understanding of the Role of
Interaction in Information Visualization

Ji Soo Yi, Youn ah Kang, John T. Stasko, Member, IEEE, and Julie A. Jacko

Abstract—Even though interaction is an important part of information visualization (InfoVis), it has garnered a relatively low level of attention from the InfoVis community. A few frameworks and taxonomies of InfoVis interaction techniques exist, but they typically focus on low-level operations and do not address the variety of benefits interaction provides. After conducting an extensive review of InfoVis systems and their interactive capabilities, we propose seven general categories of interaction techniques widely used in InfoVis: 1) Select, 2) Explore, 3) Reconfigure, 4) Encode, 5) Abstract/Elaborate, 6) Filter, and 7) Connect. These categories are organized around a user's intent while interacting with a system rather than the low-level interaction techniques provided by a system. The categories can act as a framework to help discuss and evaluate interaction techniques and hopefully lay an initial foundation toward a deeper understanding and a science of interaction.

Index Terms—information visualization, interaction, interaction techniques, taxonomy, visual analytics

1 INTRODUCTION

Information visualization (InfoVis) systems, at their core, appear to have two main components: representation and interaction. The graphics component, whose roots lie in the field of computer graphics, concerns the mapping from data to representation and how that representation is rendered on the display. The interaction component involves the dialog between the user and the system as the user explores the data set to uncover insights. The interaction component's roots lie in the area of human-computer interaction (HCI). Although discussed as two separate components, representation and interaction clearly are not mutually exclusive. For instance, interaction with a system may activate a change in representation. Nonetheless, the two components seem to compose the two fundamental aspects of InfoVis systems, and it seems reasonable to consider what each contributes to an end-user's experience.

We argue that the representation component has received the vast majority of attention in InfoVis research. A cursory scan of a recent conference proceedings or journal issues in the area will uncover many articles about new representations of data sets, but interaction is often relegated to a secondary role in these articles. Interaction rarely is the main focus of research efforts in the field, essentially making it the "little brother" of InfoVis. In other words, it is overshadowed by the more noteworthy representation aspects. A few papers have mainly focused on the interactive aspects of InfoVis (e.g., [10, 15, 25, 47]), but these are relatively uncommon when compared to papers introducing new data representations.

Interaction is an essential part of InfoVis, however. Without interaction, an InfoVis technique or system becomes a static image or autonomously animated images (e.g., InfoCarnet [23]). While static images clearly have analytic and expressive value (e.g., [8, 29, 46]), their usefulness becomes more limited as the data set that they represent grows larger with more variables. Actually, even with a static image such as a poster, a user (or a reader) will often perform several interactions (e.g., rotating the poster, looking closer further,

and jotting down notes on the poster). Spence even suggests the notion of "passive interaction" through which the user's mental model on the data set is changed or enhanced [38]. Finally, through interaction, some limits of a representation can be overcome, and the cognition of a user can be further amplified (e.g., [15, 29]).

The importance of interaction and the need for its further study seem undisputed. For example, the recent book *Illuminating the Path: The Research and Development Agenda for Visual Analytics* calls for further research on interaction:

"Recommendation 3.3: Create a new science of interaction to support visual analytics. The grand challenge of interaction is to develop a taxonomy to describe the design space of interaction techniques that supports the science of analytic reasoning. We must characterize this design space and identify under-explored areas that are relevant to visual analytics. Then, R&D should be focused on expanding the repertoire of interaction techniques that can fill those gaps in the design space." ([45], p. 76)

This recommendation concerns visual analytics which is not equivalent to InfoVis, but the two clearly share much in common and the motivation for this call can equally be applied to InfoVis.

While we believe that few would argue with the merits of the goals in the recommendation, precisely defining what is being called for is not so easy. What does it mean to create a "science of interaction" in visual analytics and InfoVis? The recommendation speaks of developing a taxonomy of interaction techniques and identifying under-explored areas for future research. These are noble efforts, but we believe that a science of interaction also should involve gaining a deeper understanding of the utility and value of interaction in these fields. What does interaction contribute to the analytic process?

For that matter, we might raise questions about the nature of interaction itself. In the context of InfoVis, what is interaction and interactive behavior? Operations such as moving a dynamic query slider [1] to narrow the set of data points being shown or selecting an alternate point in a fish-eye view [19] to change the focus seem like clear examples of interactive behavior. But consider a system where the user selects a menu operation to change from a scatter plot to a parallel coordinates of the data. Is that interaction?

The purpose of this article relates to the recommendation from *Illuminating the Path* that was discussed above. Defining a science of interaction is a lofty goal and we do not purport to do so here, but we do seek to take some initial steps toward that goal. Our objective is to further current understandings of the role that interaction plays

**Toward a Deeper Understanding of the Role of
Interaction in Information Visualization, Ji Soo
Yi, Youn ah Kang, John T. Stasko and Julie A.
Jacko, IEEE Transactions on Visualization and
Computer Graphics, (Paper presented at InfoVis
'07), Vol. 13, No. 6, November/December
2007, pp. 1224–1231.**

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Manuscript received 31 March 2007; accepted 1 August 2007; posted online 27 October 2007.
For information on obtaining reprints of this article, please send e-mail to: reprints@computer.org.

Categories of Interaction Techniques

5

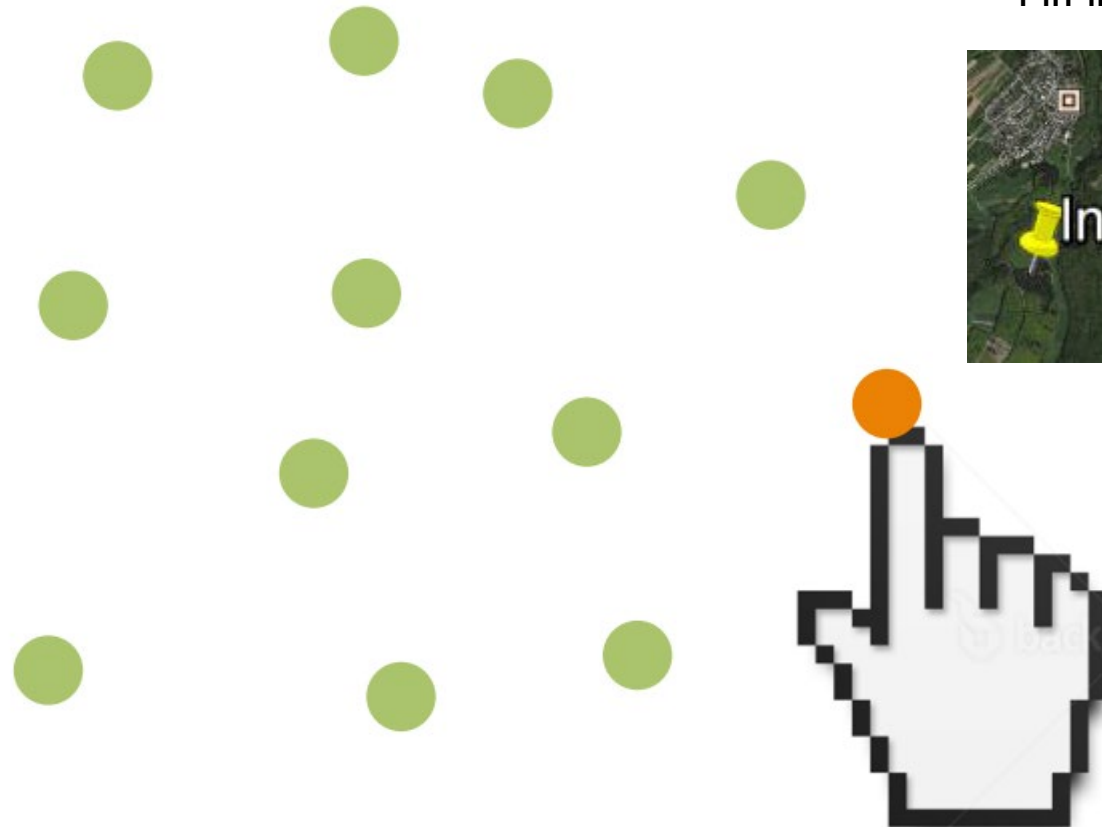
- ☐ Select
- ☐ Explore
- ☐ Encode
- ☐ Reconfigure
- ☐ Abstract/Elaborate
- ☐ Filter
- ☐ Connect

Select / Focus

6

- Mark something as interesting

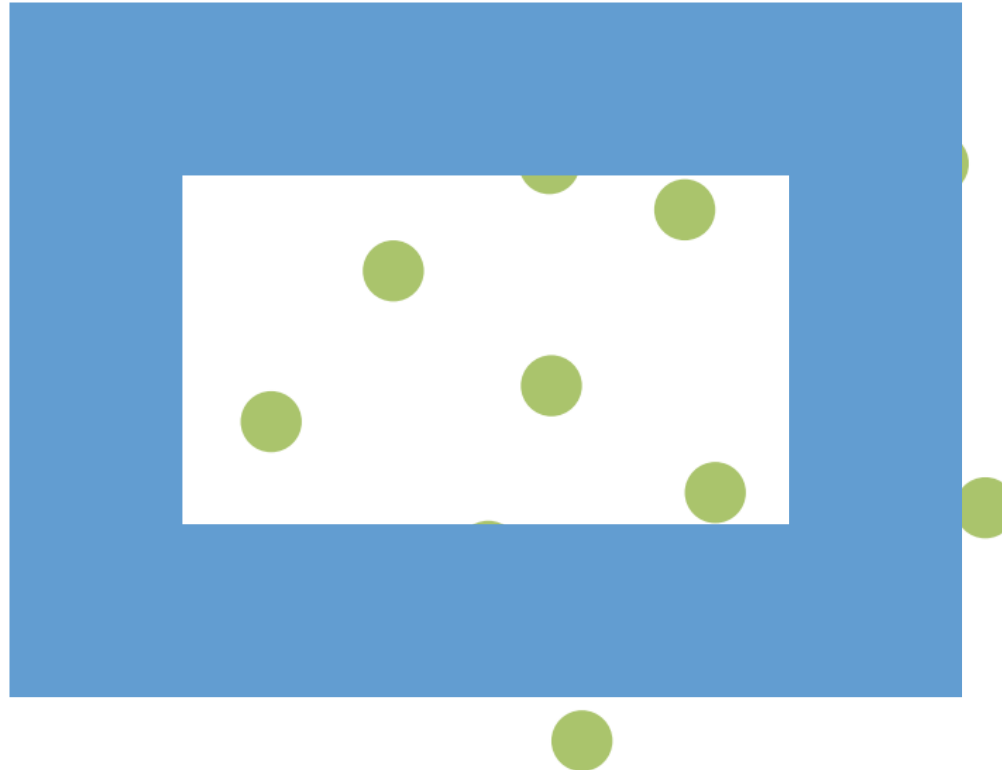
Example:
Pin in Google Earth



Explore

7

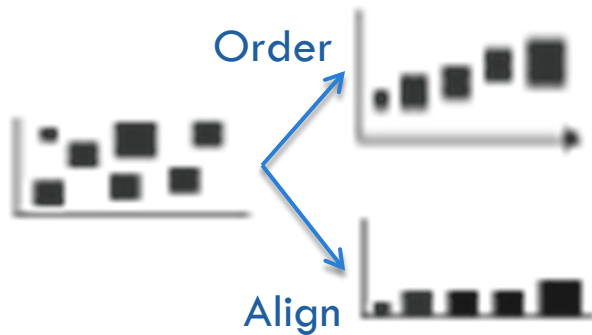
- Show me something else
 - ▣ Example: Panning (Schwenken)



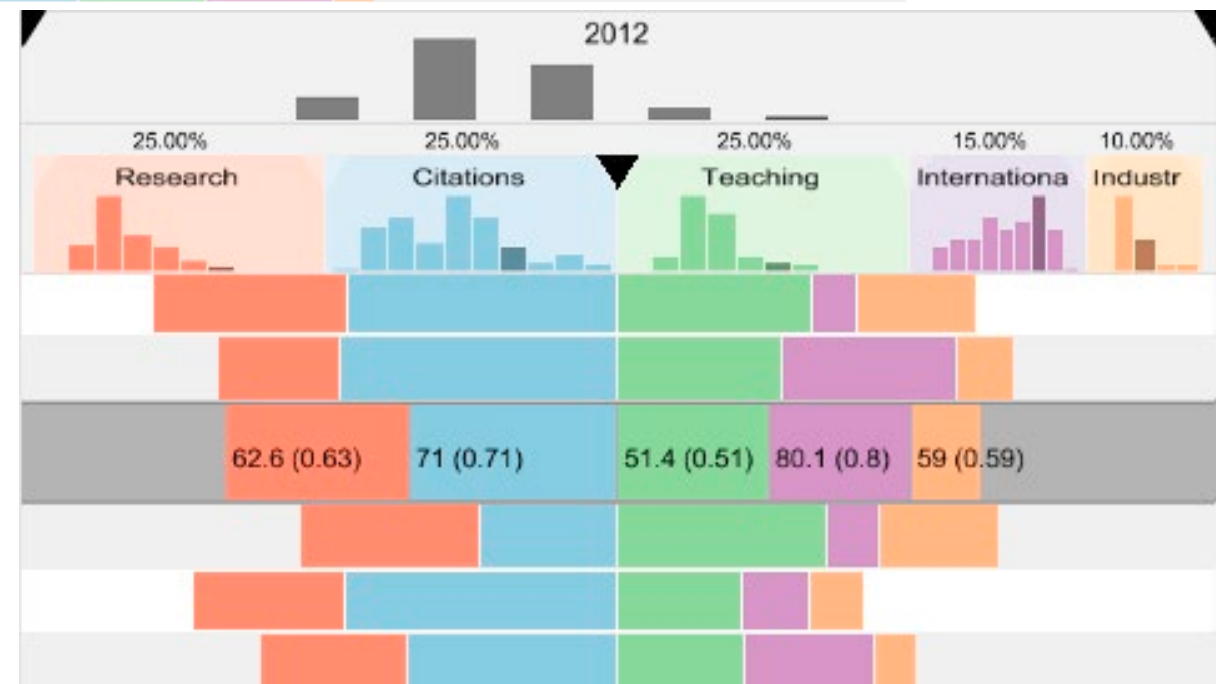
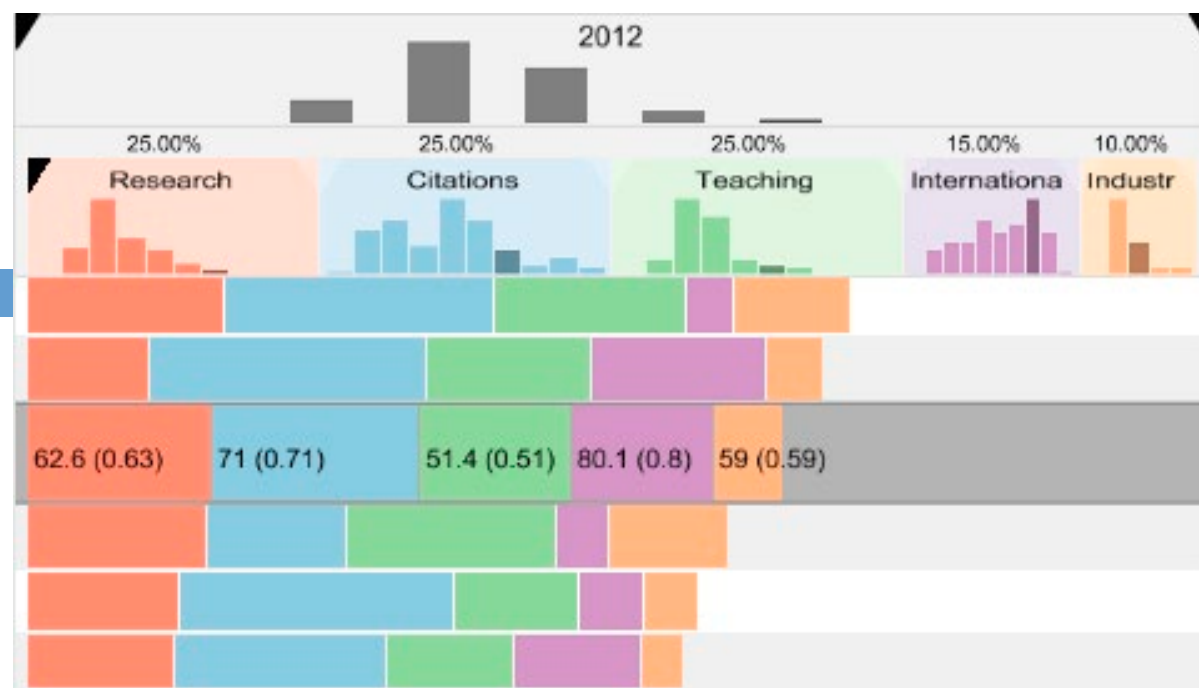
Reconfigure

8

- Change the spatial arrangement
- here: Alignment



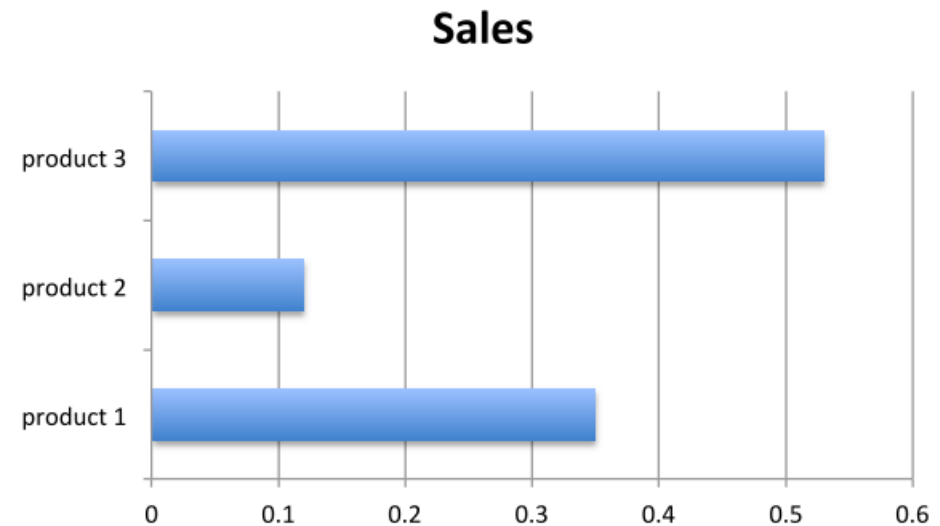
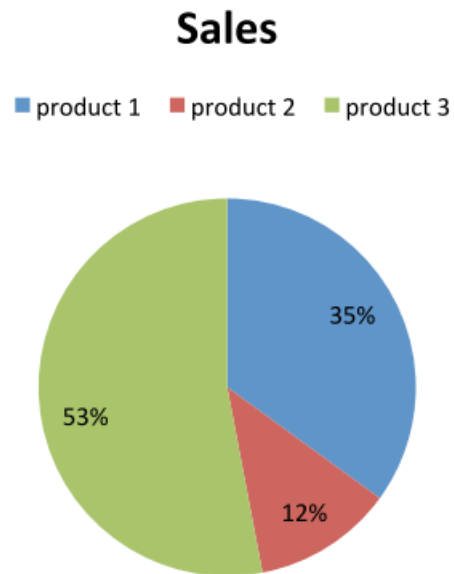
LineUp: Visual Analysis of Multi-Attribute Rankings. Gratzl, Lex, Gehlenborg, Pfister, and Streit. *IEEE Trans. Visualization and Computer Graphics (Proc. InfoVis 2013)* 19:12 (2013), 2277–2286.



Encode

9

- Show me a different visual representation
 - ▣ Example: by changing a visual variable:
 - Color, size, orientation, font, form

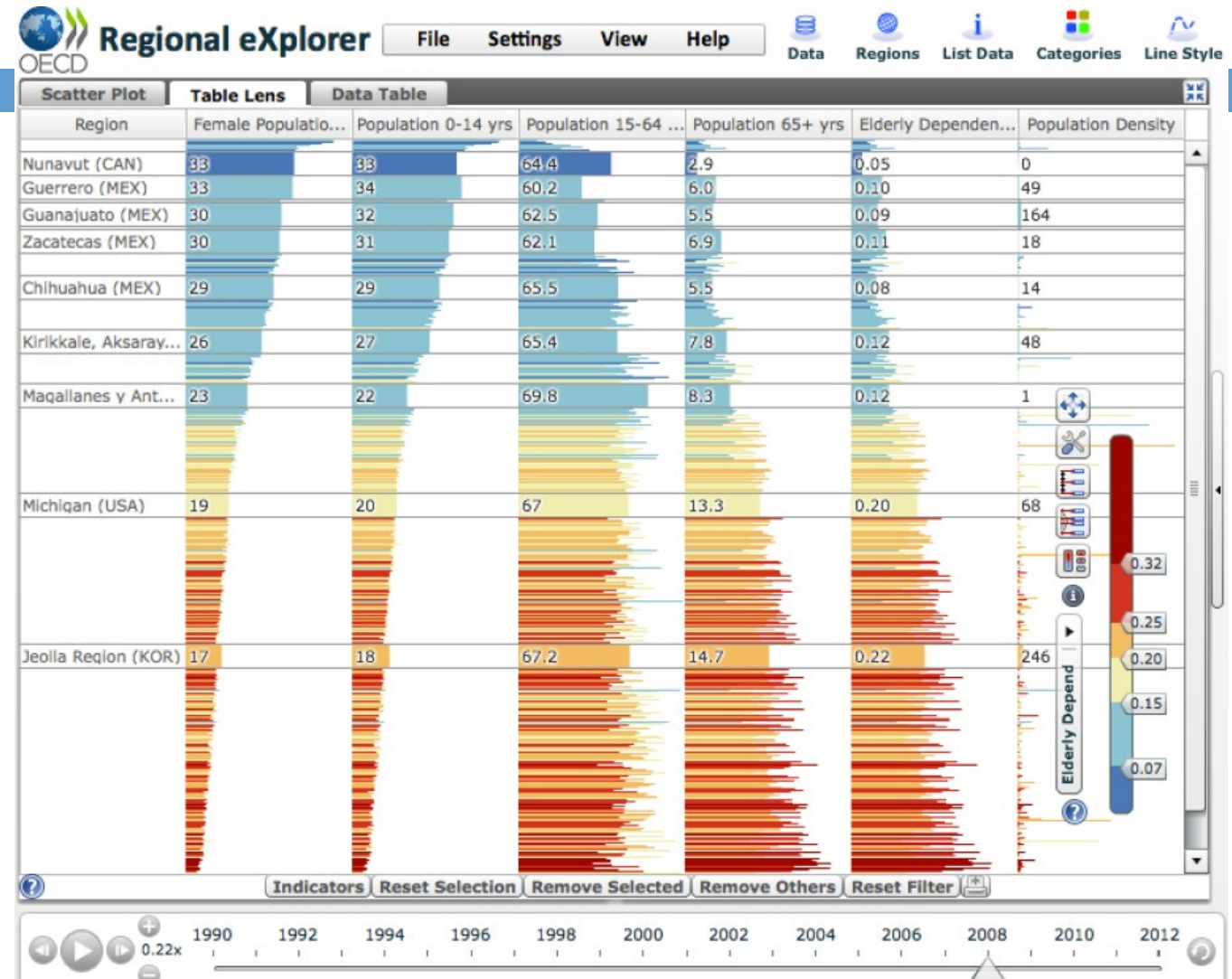


Abstract / Elaborate

10

- Show me more or less detail

Example: using Table Lens and Tooltips

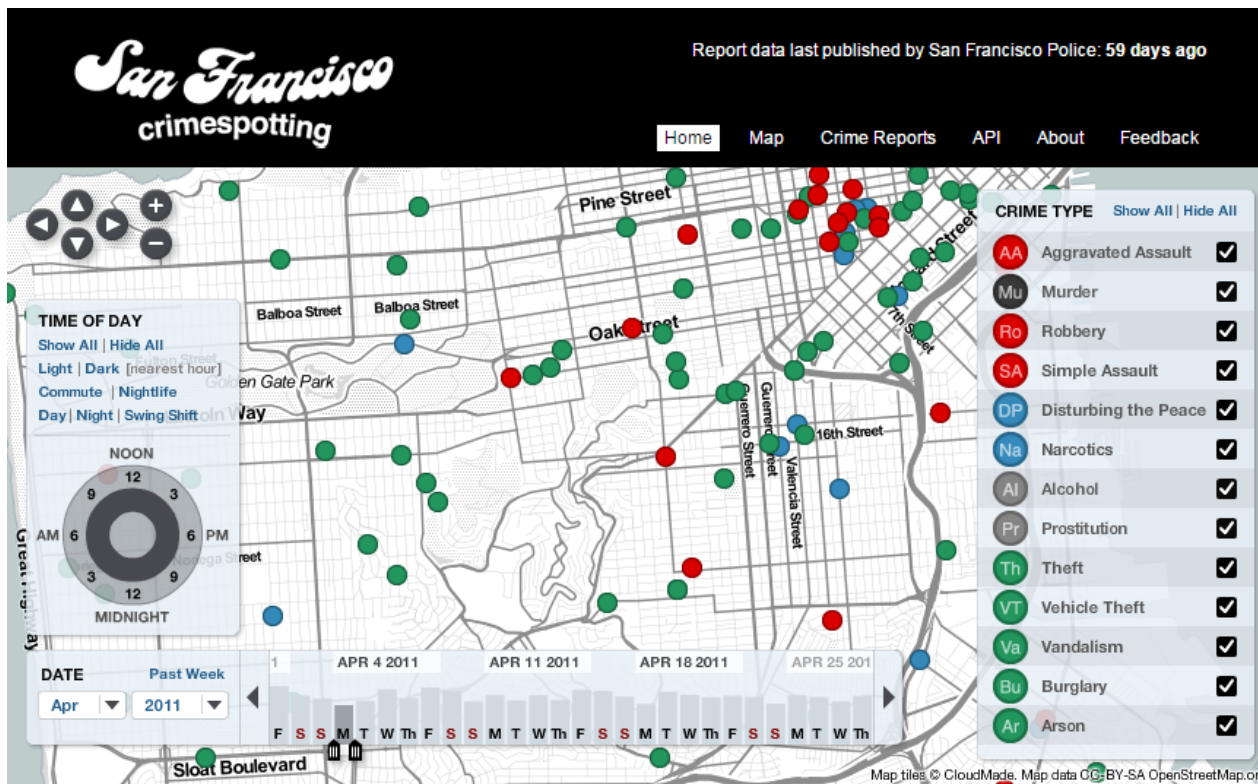


<http://stats.oecd.org/OECDregionalstatistics/#story=0>

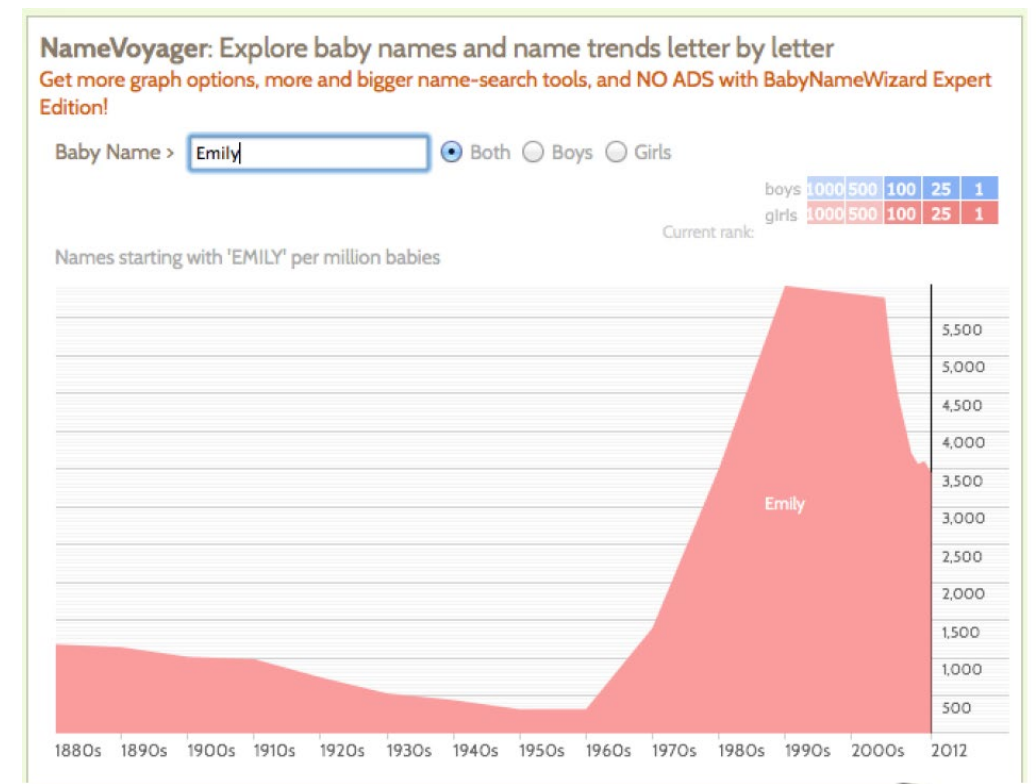
Filter

11

- Show me something conditionally (eingeschränkt/bedingt)



<http://sanfrancisco.crimespotting.org/>

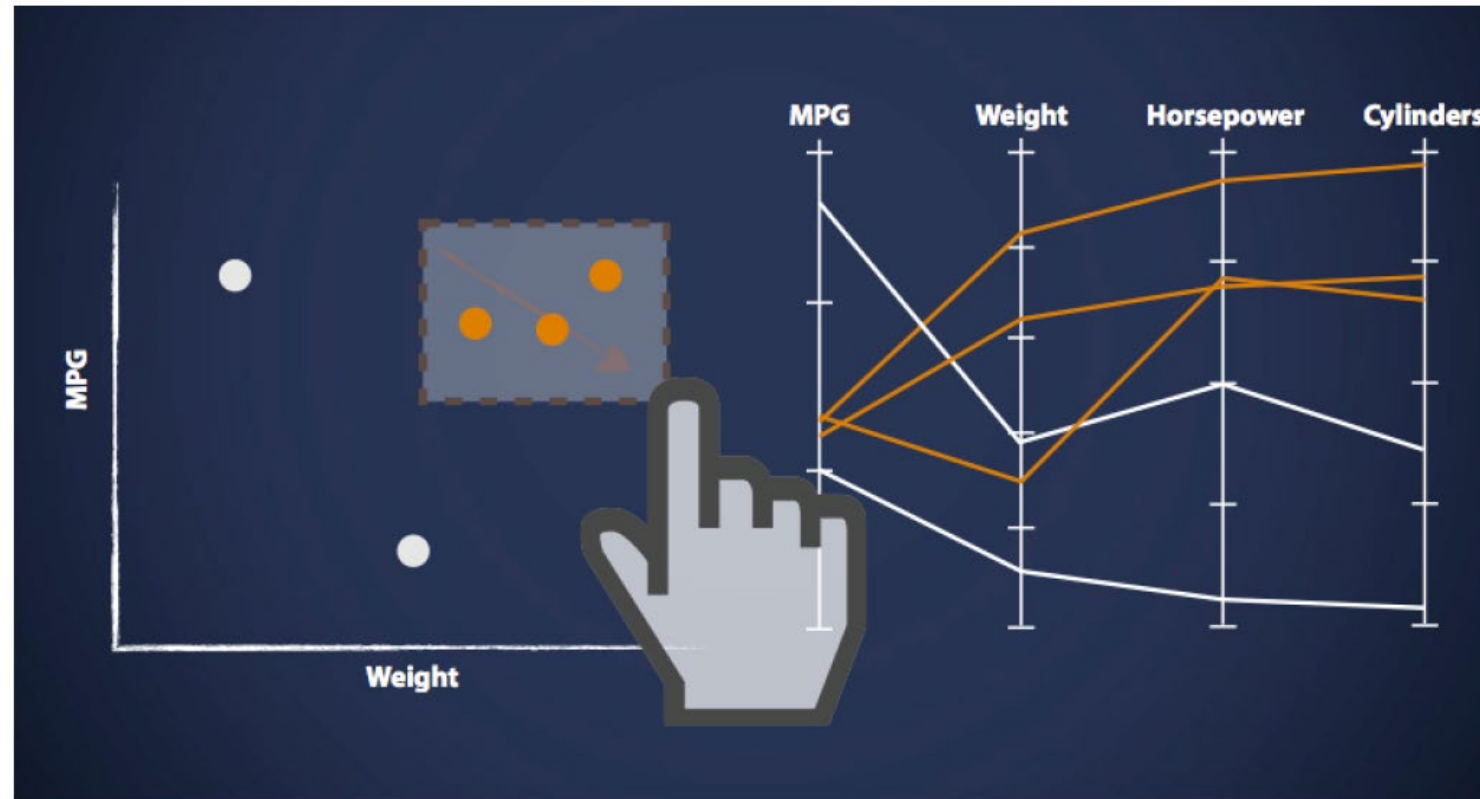


<http://www.babynamewizard.com/voyager>

Connect

12

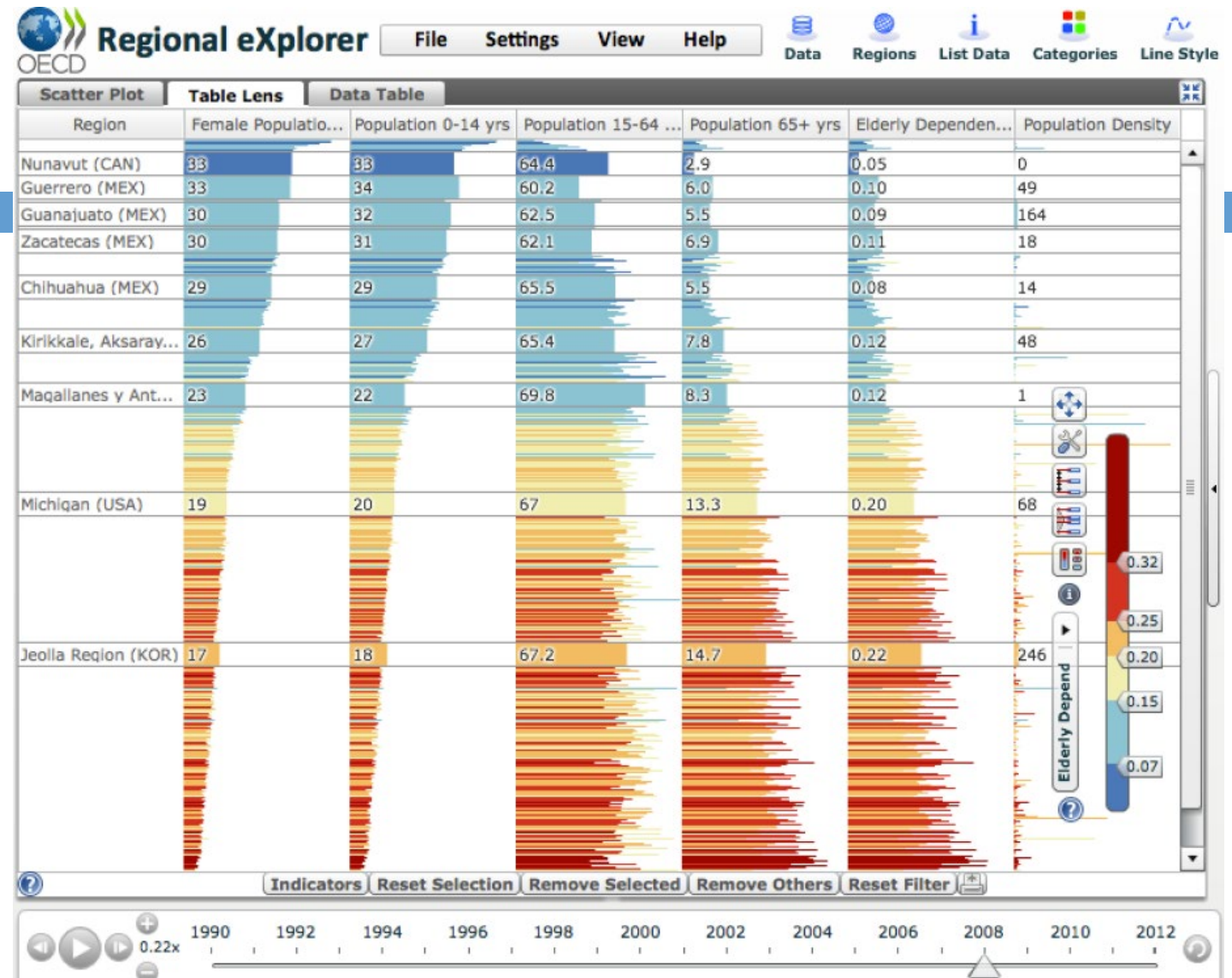
- Show me related items
 - ▣ Example: **Brushing and Linking**



Connect

13

- Brushing and Linking
 - Several views of the same data
 - Items selected in one view are highlighted in other views.



<http://stats.oecd.org/OECDregionalstatistics/#story=0>

Chapter
7.2

Other Interaction Techniques

Overview & Detail in Separate Views

15

- No distortion (Verzerrung)
- Disadvantage: continuous change of focus of attention

A Review of Overview+Detail, Zooming, and Focus+Context Interfaces. Cockburn, Karlson, and Bederson. *ACM Computing Surveys* 41:1 (2008), 1–31.



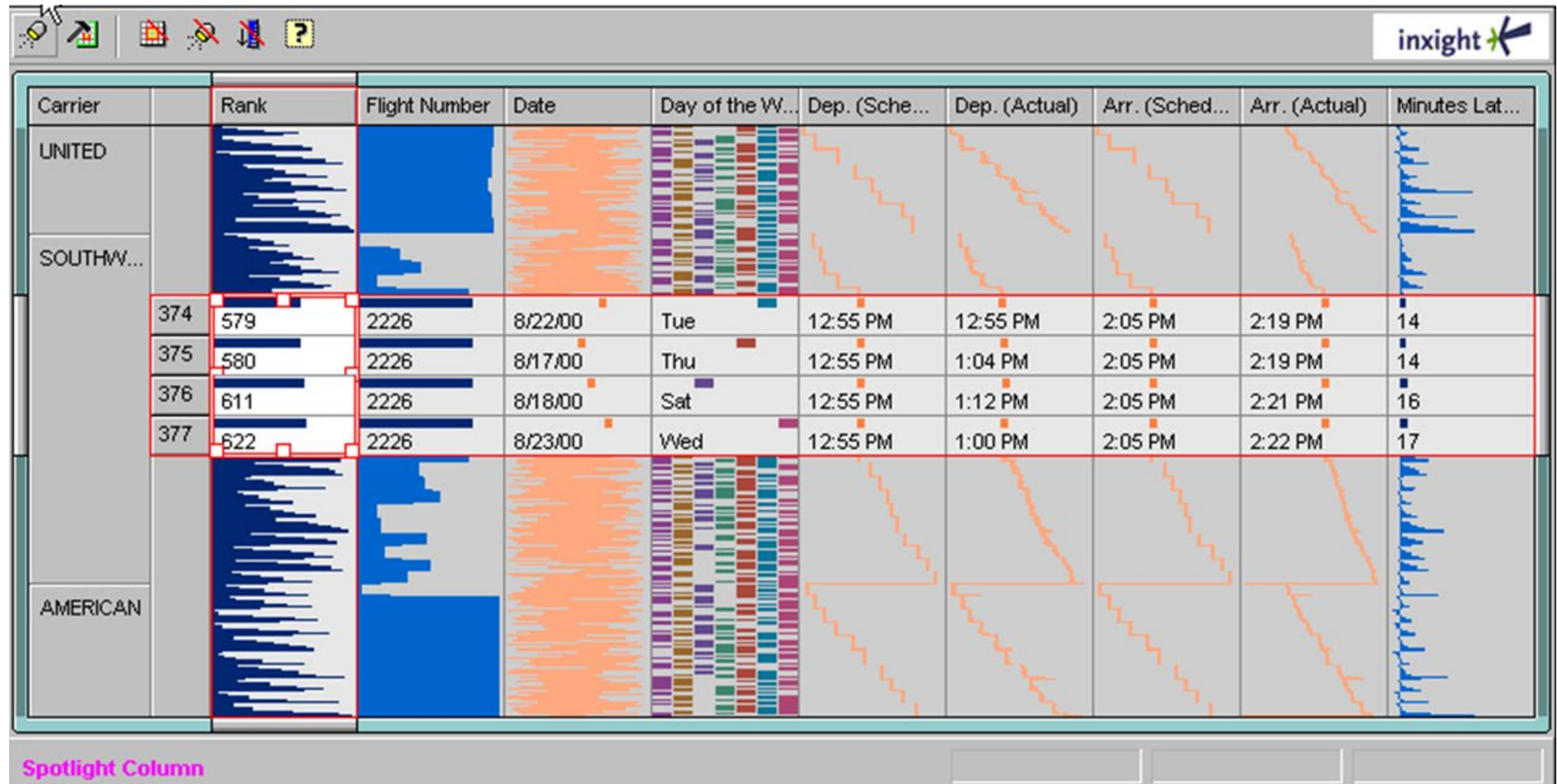
Focus & Context

16

- Both detail and context are shown **in a single view**
 - ▣ In some of the Focus&Context techniques, parts of the representation may be hard to read due to distortion.
- Examples (all developed at Xerox PARC):
 - ▣ TableLens
 - ▣ Perspective Wall
 - ▣ Hyperbolic Tree Browser

Focus & Context: TableLens

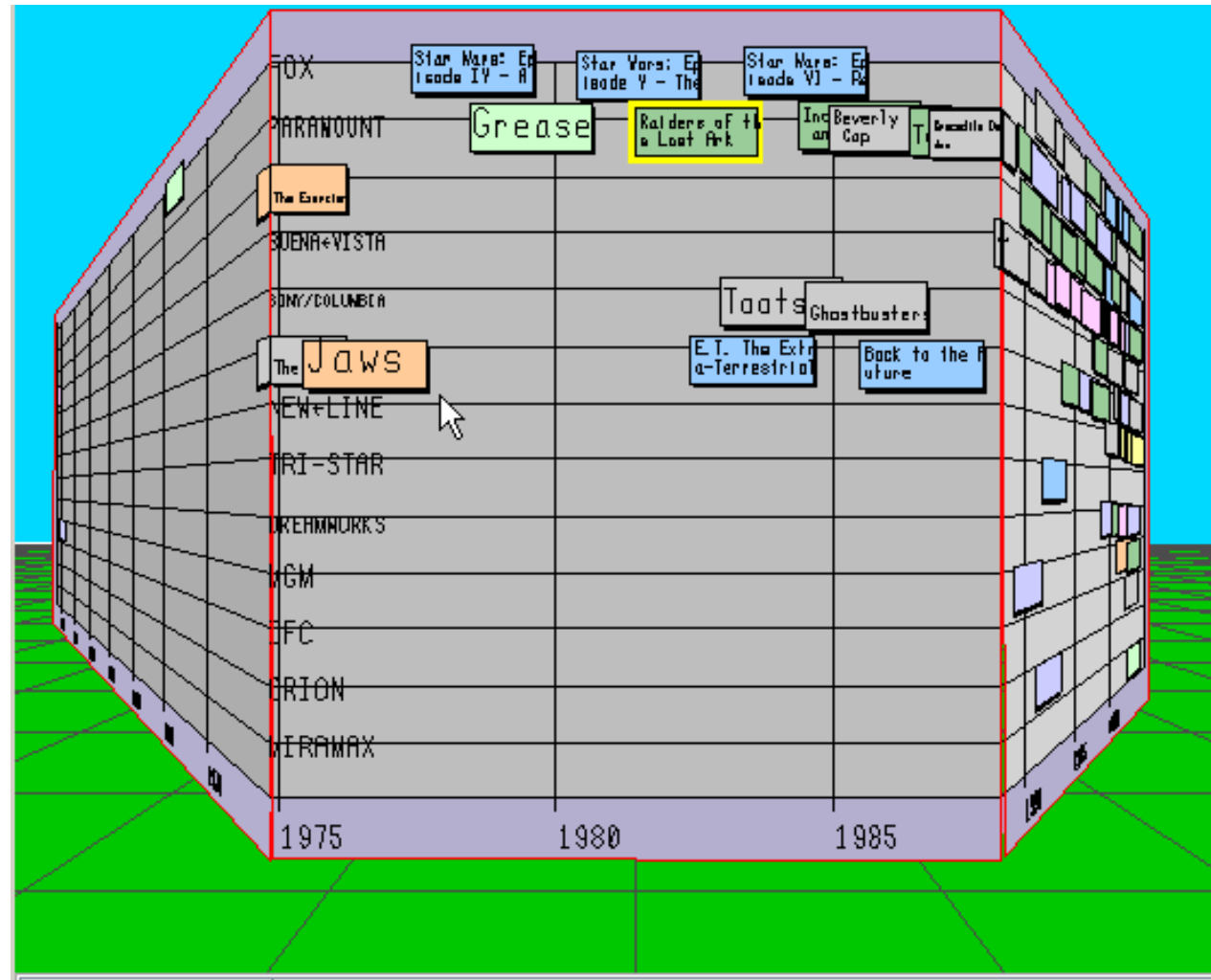
17



Focus & Context: Perspective Wall

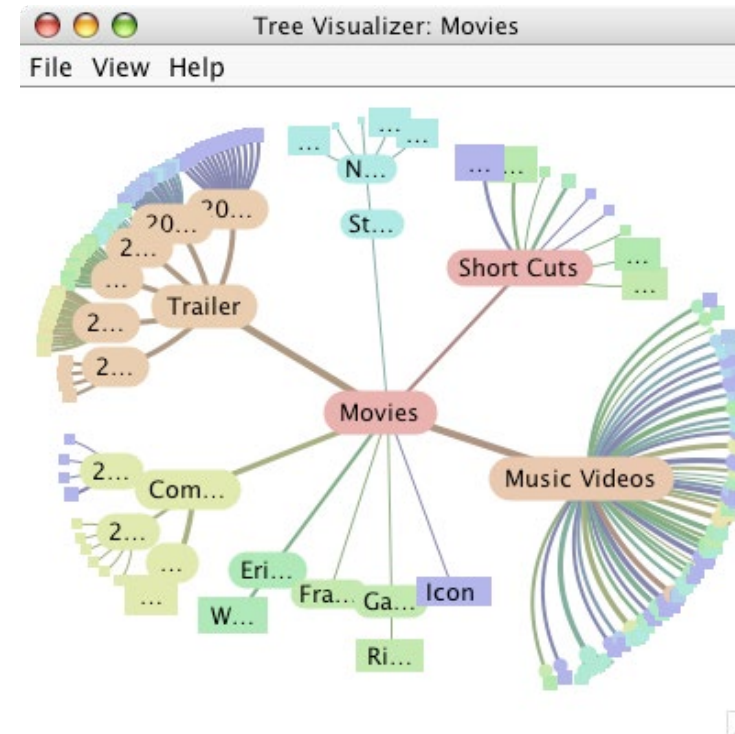
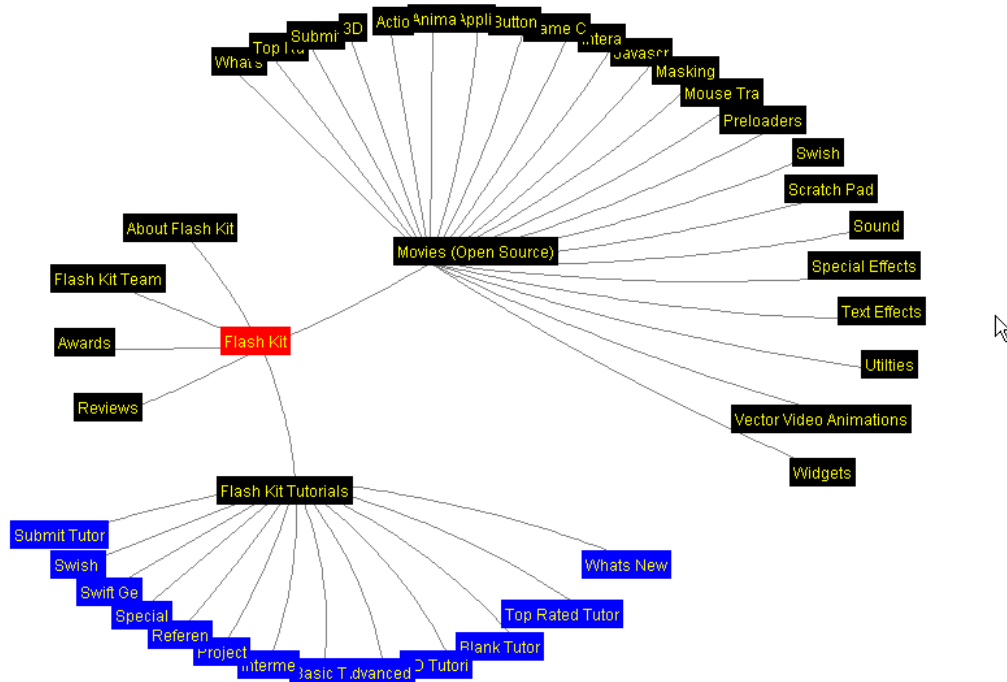
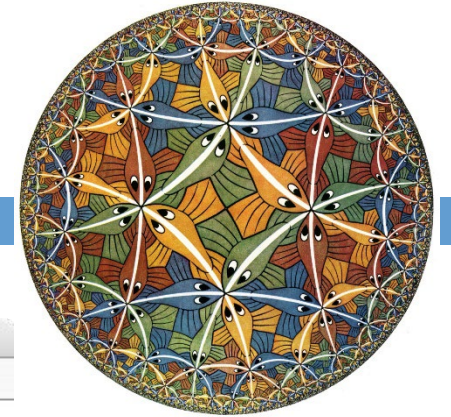
18

Example: When movies were released by different film studios.



Focus & Context: Hyperbolic Tree

19



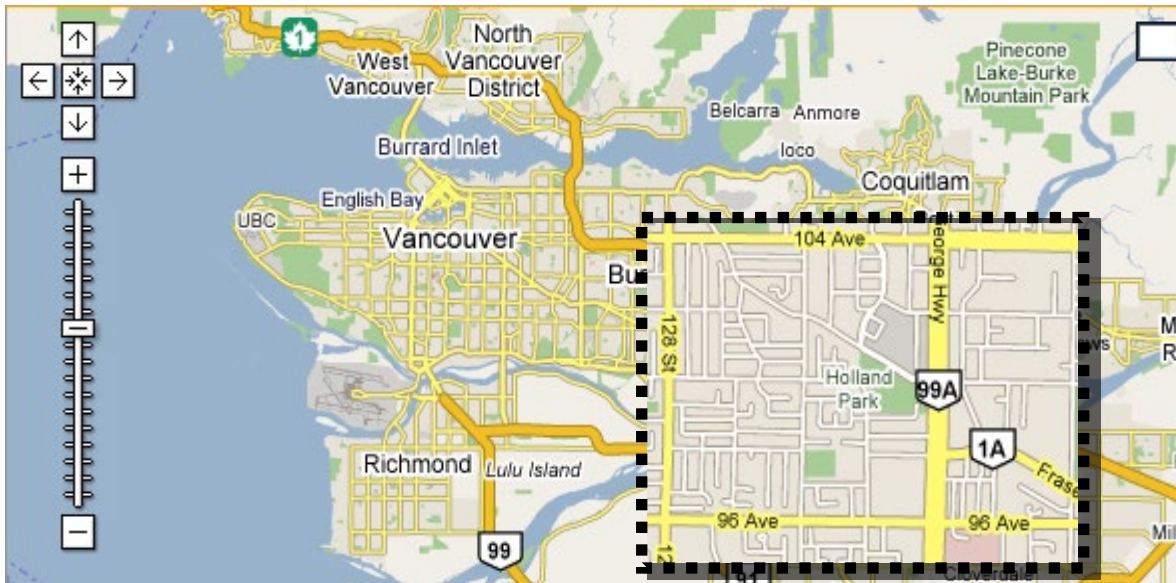
<http://philogb.github.io/jit/static/v20/Jit/Examples/Hypertree/example1.html>

<http://jowl.ontologyonline.org/HyperBolicTree.html>

<https://hyperbolic-tree-of-life.github.io/>

Focus & Context for Maps

20

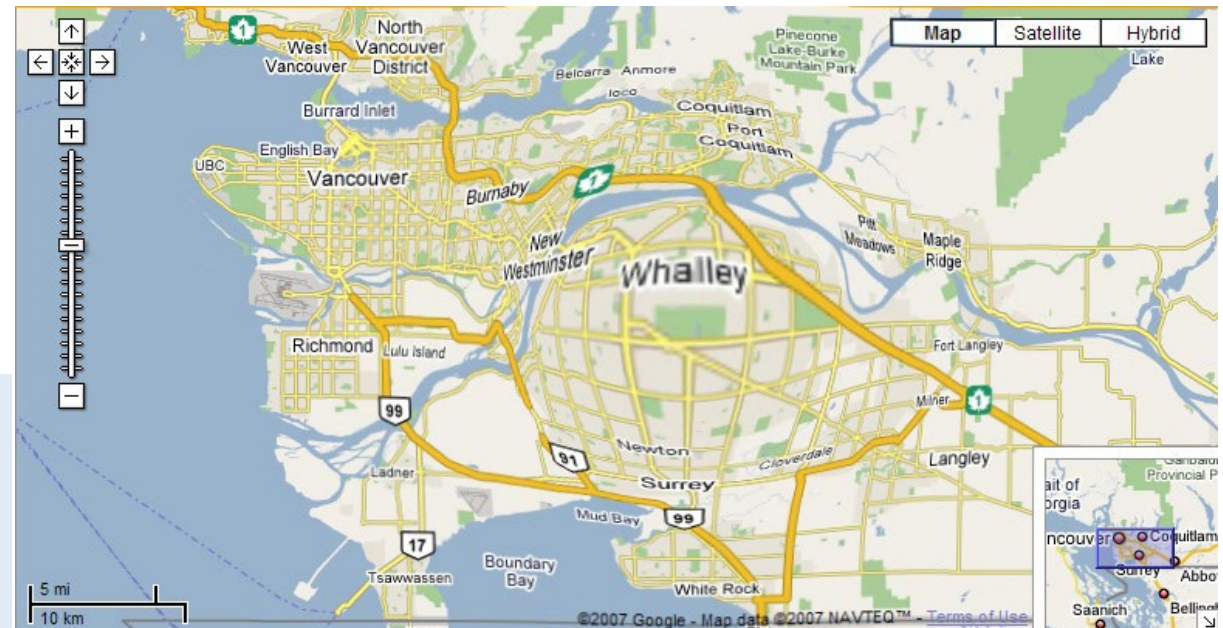


Lense:

- Enlarged image overlays the overview representation.
- Elements are hidden

Fish-eye lense:

- Continuous transition
- Distortion may be confusing

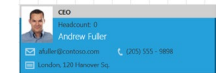
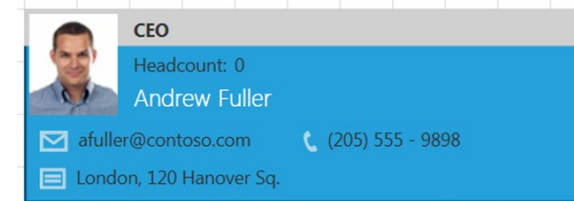


Standard vs. Semantic Zoom

21

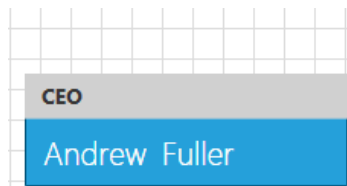
□ Geometric Zooming (Standard):

- The magnification of an image is increased or decreased.
- Example: Google Earth

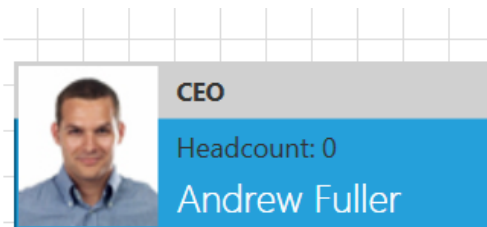


□ Semantic Zooming:

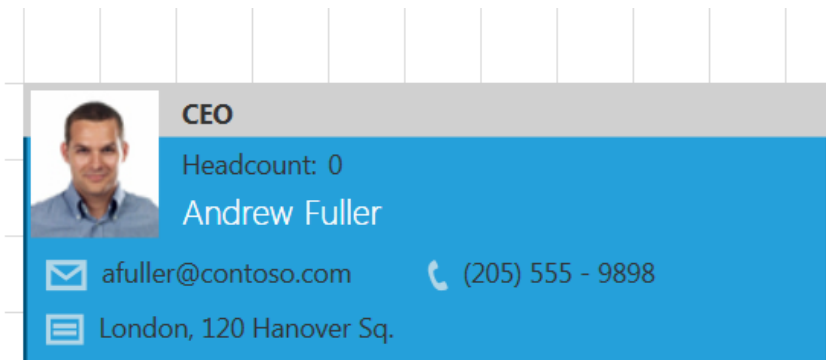
- The selection and structure of data to be displayed is changed depending on what meaning of the presented object should be preserved.



ZoomFactor = 1



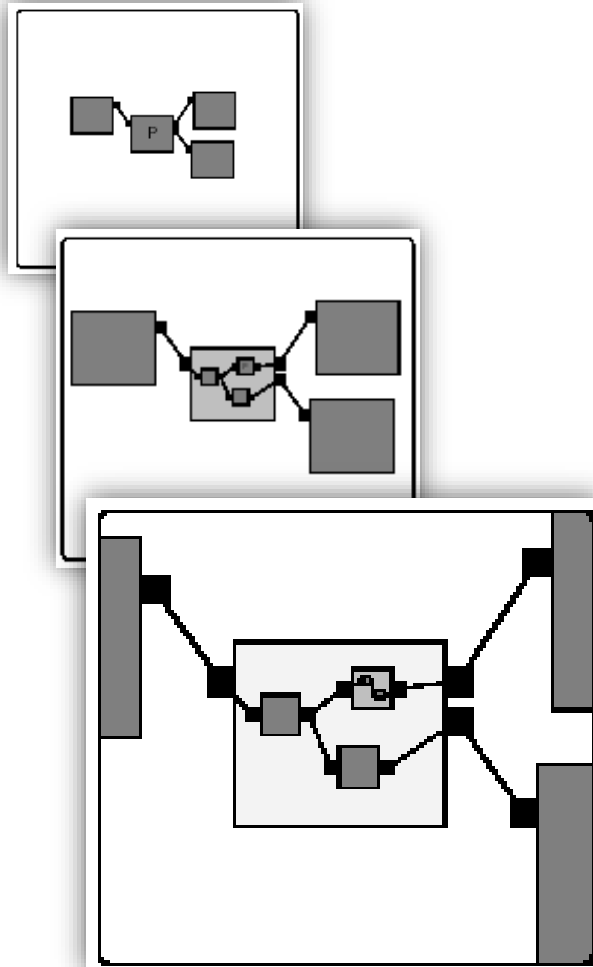
ZoomFactor = 1.5



ZoomFactor = 2.5

Semantic Zooming

22



Schedule

All schedule

My schedule

Show: Every day

Tuesday

10:00AM

12:00PM

2:30PM

4:00PM

5:45PM

7:30PM

8:30PM

Wednesday

10:00AM

12:00PM

2:00PM

2:30PM

3:30PM

5:00PM

6:30PM

8:00PM

10:30PM

Thursday

11:00AM

12:00PM

1:30PM

2:30PM

4:00PM

5:30PM

7:00PM

8:30PM

Friday

11:00AM

12:00PM

1:30PM

3:30PM

5:00PM

6:00PM

Schedule

All schedule

My schedule

Show: Every day

Thursday, 4:00PM

4:00 PM

Using tiles and notifications
Kip Olson
APP-3467 / THU / 4:00PM - 5:00PM / ACC 304.9

4:00 PM

Build data-driven collection and list ap...
Laurence Moroney
APP-8127 / THU / 4:00PM - 5:00PM / MARBOTT...

4:00 PM

Certifying graphics experiences on Wi...
Rudolph Balaz
WIN-2157 / THU / 4:00PM - 5:00PM / MARBOTT...

4:00 PM

Moving driver quality upstream with...
Jakob Lichtenberg, Khuram Shahid
WIN-2287 / THU / 4:00PM - 5:00PM / MARBOTT...

4:00 PM

Running Windows from an external US...
Arvind Padole, Steve Silverberg
WIN-3167 / THU / 4:00PM - 5:00PM / MARBOTT...

4:00 PM

Windows Certification: improvements...
Oscar Koenders
WIN-2287 / THU / 4:00PM - 5:00PM / ACC 304.9

4:00 PM

Building and delivering a great Metro...
Max Morris
WIN-2757 / THU / 4:00PM - 5:00PM / MARBOTT...

4:00 PM

Understanding Wi-Fi Direct in Window...
Mitesh Desai, Yatharth Gupta
WIN-3297 / THU / 4:00PM - 5:00PM / MARBOTT...

4:00 PM

Building Windows runtime sockets apps
Dave Thaler, Peter Smith
PLAT-1507 / THU / 4:00PM - 5:00PM / MARBOTT...

4:00 PM

Building Metro style apps that print
Daren Davis, Jon Kay
PLAT-6797 / THU / 4:00PM - 5:00PM / MARBOTT...

4:00 PM

3D Graphics in Metro Style Apps and...
Chas Boyd
PLAT-7527 / THU / 4:00PM - 5:00PM / MARBOTT...

4:00 PM

Capturing personal photos, video, and...
Khuram Zia, Mehmet Kuucugoz
PLAT-7777 / THU / 4:00PM - 5:00PM / ACC 304.9

4:00 PM

Designing Metro style apps using CSS3
Christian Stockwell
PLAT-8707 / THU / 4:00PM - 5:00PM / ACC 310

4:00 PM

Building great Metro style gallery app...
Ed Averett
PLAT-8927 / THU / 4:00PM - 5:00PM / ACC 304.9

4:00 PM

Using classification for access control...
Matthias Wolnik
SAC-4287 / THU / 4:00PM - 5:00PM / ACC 304.9

4:00 PM

Design for a multimillion thin client m...
Karthik Lakshminarayanan, Rob Williams
SAC-4287 / THU / 4:00PM - 5:00PM / MARBOTT...

4:00 PM

Enabling multi-tenancy and converg...
Charley Wen, Richard Wurdack
SAC-4287 / THU / 4:00PM - 5:00PM / ACC 304.9

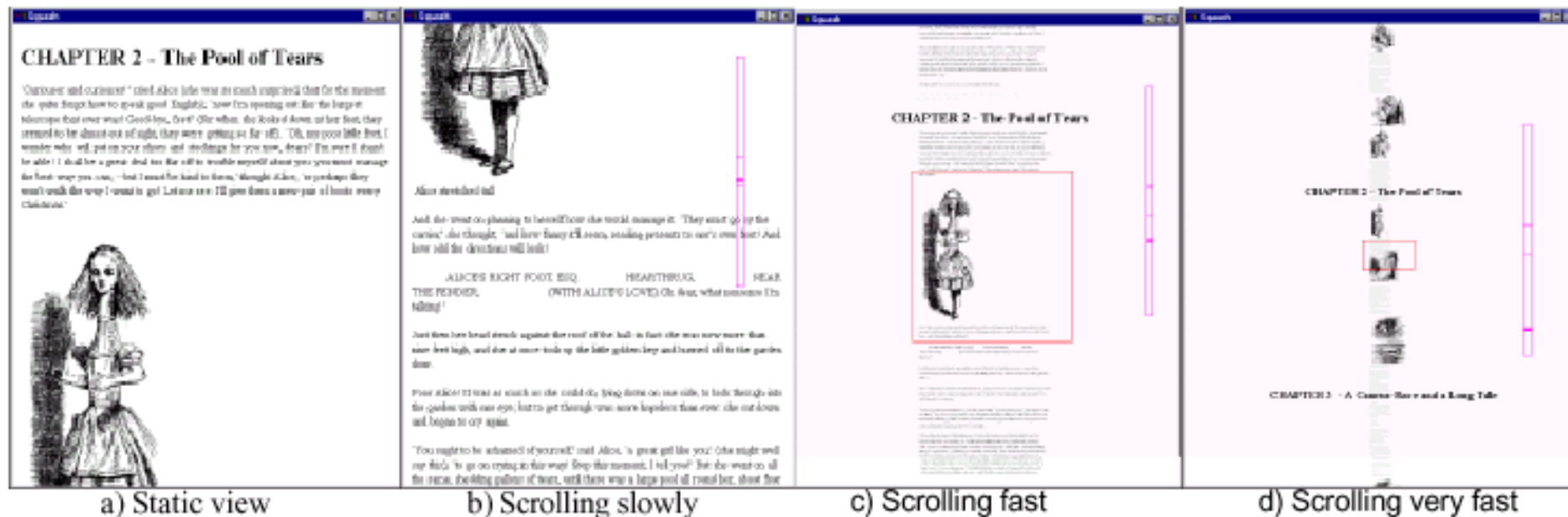
4:00 PM

Designing systems for continuous...
Jim Pinkerton, Surendra Verma
SAC-4287 / THU / 4:00PM - 5:00PM / ACC 304.9

Speed-dependent Zooming

23

<http://www-ui.is.s.u-tokyo.ac.jp/~takeo/research/autozoom/autozoom.htm>



Speed-dependent automatic zooming for browsing large documents, Takeo Igarashi, Ken Hinckley , 13th Annual Symposium on User Interface Software and Technology, ACM UIST'00, San Diego, CA , November 5-8, 2000, pp.139-148.

Shneiderman's Taxonomy of Information Visualization Tasks

24

- **Overview:** see overall patterns, trends
- **Zoom:** see a smaller subset of the data
- **Filter:** see a subset based on values, etc.
- **Details on demand:** see values of objects when interactively selected
- **Relate:** see relationships, compare values
- **History:** keep track of actions and insights
- **Extract:** mark and capture data

"The Mantra appears to act as an inspiration and guideline for practitioners rather than a scientifically studied methodological approach."
[Craft&Cairns, IV 2005]



Shneiderman's Information-Seeking Mantra:
Overview first, zoom & filter, then details-on-demand

Chapter
7.3

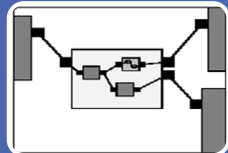
Summary



Interaction techniques: Select, Explore, Encode, Reconfigure, Abstract/Elaborate, Filter, Connect



Overview & Detail vs. Focus & Context



Geometric vs. Semantic Zooming



Shneiderman's Information-Seeking Mantra: Overview first, zoom & filter, then details-on-demand